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AI LOAN ELIGIBILITY CHECKER — PROJECT DOCUMENTATION

1. Cover Page

- **Project Name:** AI Loan Eligibility Checker
 - Short tagline
 - Team/member names
 - College/institution
 - Date
 - Project logo if you have one
-

2. Project Overview

Briefly explain:

AI Loan Eligibility Checker is a web-based financial decision-support application that analyzes user-provided financial information and provides an estimated loan eligibility result along with supporting financial insights.

Also mention the purpose and what the application provides.

3. Problem Statement

Explain the problem:

- Loan eligibility can depend on multiple financial factors.
- Users may not understand how these factors affect eligibility.

- Manual evaluation can be difficult for users to understand.
 - Users need a simple way to analyze their financial information before applying.
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4. Proposed Solution

Explain how **your system** solves it.

User enters financial information



System validates the information



Data is processed



Eligibility model analyzes the data



Eligibility result is generated



User receives explanation + financial insights

5. Objectives

Your objectives can include:

- Build an easy-to-use loan eligibility platform.
- Analyze important financial parameters.
- Provide an estimated eligibility result.
- Explain the factors affecting the result.
- Provide EMI calculations.

- Provide credit-related insights.
 - Provide AI-generated financial tips.
 - Store user information/results where appropriate.
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6. Target Users

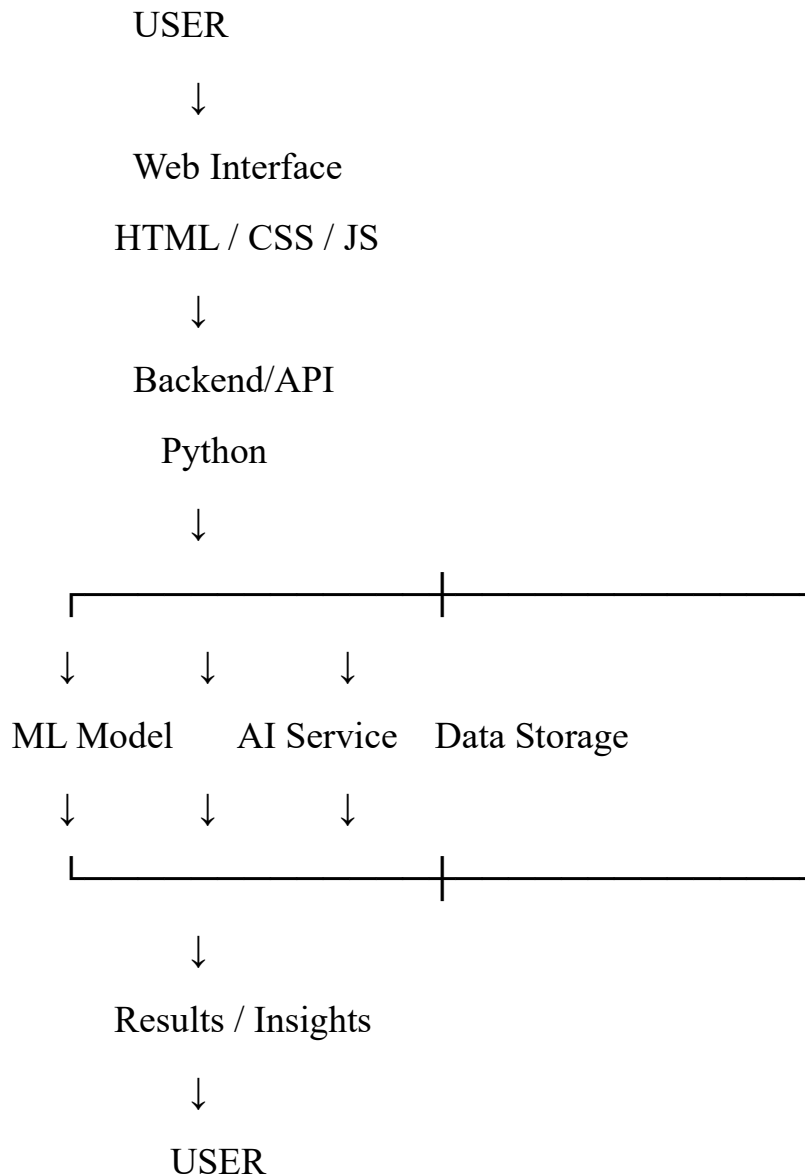
For example:

- Students/young adults learning about personal finance
 - Salaried individuals
 - People planning to apply for loans
 - Users who want to understand their financial position
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7. System Architecture

This is important.

Show the complete technical structure:



You can make this into an actual visual diagram in the PDF.

8. Complete User Workflow

This should show what happens from opening the website to getting the result.

Start



Open Website



Login / Sign Up



Dashboard



Select Loan Eligibility Checker



Enter Financial Details



Validate Input



Process Data



Run Eligibility Model



Generate Result



Show Explanation



Show Financial Tips



Save Result



End

9. Main Features

This is one of the biggest sections.



Feature 1 — Loan Eligibility Checker

Explain:

- User enters financial information.
- System processes the information.
- Model evaluates the provided factors.
- System generates an estimated eligibility result.
- User receives an explanation of the result.

Possible inputs:

- Monthly/annual income
 - Employment information
 - Loan amount
 - Loan tenure
 - Existing loans
 - Credit score
 - Other relevant financial information
-



Feature 2 — Credit Score Analyzer

Explain:

Allows users to enter their credit score and understand its significance in the context of loan applications.

Could display:

- Credit score
 - General interpretation
 - Factors that can affect creditworthiness
 - Suggestions for responsible financial management
-

Feature 3 — EMI Calculator

Workflow:

Loan Amount

+

Interest Rate

+

Loan Tenure

↓

EMI Calculation

↓

Monthly EMI

↓

Total Interest

↓

Total Repayment

Feature 4 — AI Financial Tips

The AI can provide general educational guidance based on the information entered by the user.

For example:

- Managing existing debt
- Understanding EMI burden
- Improving financial planning
- Preparing before applying for a loan

Clearly state that these are **general financial insights, not guaranteed financial advice or loan approval decisions.**

Feature 5 — User Account

Include:

Sign Up



Login



Dashboard



Access Tools



View Previous Results

Feature 6 — Data Storage

Explain what information is stored and why.

For your planned project, you can document:

Google Sheets → user/project data persistence

Be specific about what you actually store once implemented.

10. Input → Processing → Output

This makes the project very easy to understand.

Input	Processing	Output
Income	Data processing	Eligibility result
Credit score	Model analysis	Credit insight
Loan amount	Calculation	EMI
Loan tenure	Calculation	Repayment information
Existing obligations	Analysis	Financial insight

11. Technology Stack

Frontend

- HTML5
- CSS3
- JavaScript

Backend

- Python
- FastAPI/Flask — **whichever you actually use**

AI

- Claude API

Machine Learning

- Your selected ML algorithm/model

Data

- Google Sheets

Deployment

- Netlify for frontend, if that's what you actually deploy with
- Backend hosting separately if applicable

Don't list technologies you haven't actually used yet.

12. AI/ML Workflow

A separate diagram would be useful:

Financial Dataset



Data Cleaning



Feature Selection



Data Preparation



Model Training



Model Testing



Prediction



Loan Eligibility Result

Then explain each step briefly.

13. Data Flow Diagram

Level 0

User



AI Loan Eligibility

System



Eligibility Result



User

Level 1

User



Enter Details



Input Validation



Data Processing



Eligibility Prediction



Result Generation



Database / Storage



Dashboard

14. UI / Page Structure

Document every page.

Landing Page

- Project introduction
- Features
- CTA / Get Started

Login Page

- Email/username
- Password
- Login

Signup Page

- User registration
- Account creation

Dashboard

- Loan Eligibility
- Credit Analyzer

- EMI Calculator
- AI Financial Tips
- Previous results

Loan Eligibility Page

- Financial input form
- Calculate button
- Result section

Result Page

- Eligibility result
 - Key factors
 - Explanation
 - Financial insights
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15. Database / Data Structure

Show what kind of information your system handles.

Example:

User

└─ User ID

└─ Name

└─ Email

└─ Account information

Financial Data

└─ Income

- |— Credit Score
- |— Loan Amount
- |— Loan Tenure
- └— Existing Obligations

Result

- |— Eligibility Result
- |— Important Factors
- └— Timestamp

Only include fields you actually plan to implement.

16. Security & Privacy

Very important for a financial project.

Mention:

- User authentication
 - Password protection/hashing if implemented
 - Input validation
 - Secure API handling
 - Avoid exposing API keys in frontend code
 - Responsible handling of financial information
 - Minimum necessary data collection
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17. Limitations

Be honest.

Examples:

- The system does not guarantee actual loan approval.
 - Results depend on the quality of the information provided.
 - A model may not represent every lender's actual criteria.
 - Real lending decisions may consider additional factors.
 - AI-generated tips should not replace professional financial advice.
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18. Testing

Include a small table:

Test	Expected Result	Status
Valid financial inputs	Result generated	✓
Empty required field	Validation message	✓
Invalid value	Error shown	✓
EMI calculation	Correct calculation	✓
Login	User authenticated	✓
API connection	Response received	✓

Update the status based on your **actual testing**.

19. Future Scope

Possible future additions:

- More sophisticated ML models
- Real-time financial data
- Additional loan types

- Personalized financial dashboards
 - Mobile application
 - More advanced credit analysis
 - Improved explainability
 - Integration with financial institutions
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20. Conclusion

Shortly summarize:

The AI Loan Eligibility Checker combines web development, data processing, machine learning and AI to create a user-friendly financial decision-support platform. It helps users understand their financial information and explore loan-related outcomes through a single interface.

21. References

At the end:

- Technologies/documentation used
- ML libraries
- API documentation
- Dataset source
- Research papers/articles used